

Group by & Aggregate Function

In [2]: `import pandas as pd`

In [4]: `df = pd.read_csv(r"Flavors.csv")`
`df`

Out[4]:

	Flavor	Base Flavor	Liked	Flavor Rating	Texture Rating	Total Rating
0	Mint Chocolate Chip	Vanilla	Yes	10.0	8.0	18.0
1	Chocolate	Chocolate	Yes	8.8	7.6	16.6
2	Vanilla	Vanilla	No	4.7	5.0	9.7
3	Cookie Dough	Vanilla	Yes	6.9	6.5	13.4
4	Rocky Road	Chocolate	Yes	8.2	7.0	15.2
5	Pistachio	Vanilla	No	2.3	3.4	5.7
6	Cake Batter	Vanilla	Yes	6.5	6.0	12.5
7	Neapolitan	Vanilla	No	3.8	5.0	8.8
8	Chocolte Fudge Brownie	Chocolate	Yes	8.2	7.1	15.3

In [10]: `df.groupby('Base Flavor').mean(numeric_only=True) # group & take out mean`

Out[10]:

	Flavor Rating	Texture Rating	Total Rating
Base Flavor			
Chocolate	8.4	7.233333	15.70
Vanilla	5.7	5.650000	11.35

In [11]: `df.groupby('Base Flavor').count() # telling number of occurence`

Out[11]:

	Flavor	Liked	Flavor Rating	Texture Rating	Total Rating
Base Flavor					
Chocolate	3	3	3	3	3
Vanilla	6	6	6	6	6

In [12]: `df.groupby('Base Flavor').sum() # summing up column`

Out[12]:

	Flavor	Liked	Flavor Rating	Texture Rating	Total Rating
Base Flavor					
Chocolate	ChocolateRocky RoadChocolte Fudge Brownie	YesYesYes	25.2	21.7	47.1
Vanilla	Mint Chocolate ChipVanillaCookie DoughPistachi...	YesNoYesNoYesNo	34.2	33.9	68.1

In [17]:

df

Out[17]:

	Flavor	Base Flavor	Liked	Flavor Rating	Texture Rating	Total Rating
0	Mint Chocolate Chip	Vanilla	Yes	10.0	8.0	18.0
1	Chocolate	Chocolate	Yes	8.8	7.6	16.6
2	Vanilla	Vanilla	No	4.7	5.0	9.7
3	Cookie Dough	Vanilla	Yes	6.9	6.5	13.4
4	Rocky Road	Chocolate	Yes	8.2	7.0	15.2
5	Pistachio	Vanilla	No	2.3	3.4	5.7
6	Cake Batter	Vanilla	Yes	6.5	6.0	12.5
7	Neapolitan	Vanilla	No	3.8	5.0	8.8
8	Chocolte Fudge Brownie	Chocolate	Yes	8.2	7.1	15.3

In [13]:

df.groupby('Base Flavor').agg({'Flavor Rating': ['mean','max','count','sum'], 'Text

Out[13]:

	Flavor Rating				Texture Rating			
	mean	max	count	sum	mean	max	count	sum
Base Flavor								
Chocolate	8.4	8.8	3	25.2	7.233333	7.6	3	21.7
Vanilla	5.7	10.0	6	34.2	5.650000	8.0	6	33.9

In [14]:

df.groupby(['Base Flavor','Liked']).agg({'Flavor Rating': ['mean','max','count','su

Out[14]:

		Flavor Rating			
		mean	max	count	sum
Base Flavor	Liked				
Chocolate	Yes	8.4	8.8	3	25.2
Vanilla	No	3.6	4.7	3	10.8
	Yes	7.8	10.0	3	23.4

```
In [16]: df.groupby('Base Flavor').describe() # all count, mean, min, max, std
```

Out[16]:

Base Flavor	Flavor Rating								Texture			
	count	mean	std	min	25%	50%	75%	max	count	mean	...	75%
Chocolate	3.0	8.4	0.346410	8.2	8.200	8.2	8.5	8.8	3.0	7.233333	...	7.35
Vanilla	6.0	5.7	2.710719	2.3	4.025	5.6	6.8	10.0	6.0	5.650000	...	6.37

2 rows × 24 columns

